

Movie Recommender System

Hybrid Approach using MovieLens Dataset

Data Science Project (Phase 4)

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Introduction

- ▶ Recommendation systems help users find relevant content from large datasets
- ▶ Used in platforms like Netflix, Amazon and Spotify
- ▶ Improve user experience by personalizing suggestions.

Problem Statement

- ▶ Users struggle to find relevant movies due to large content volume
- ▶ Need for a system that gives personalized movie recommendation
- ▶ Goal: build an intelligent movie recommender system

Project Objective

- ▶ Build a hybrid movie recommendation system
- ▶ Use MovieLens dataset (ratings, movies, tags)
- ▶ Combine
 - content-based filtering
 - collaborative filtering

Dataset Overview

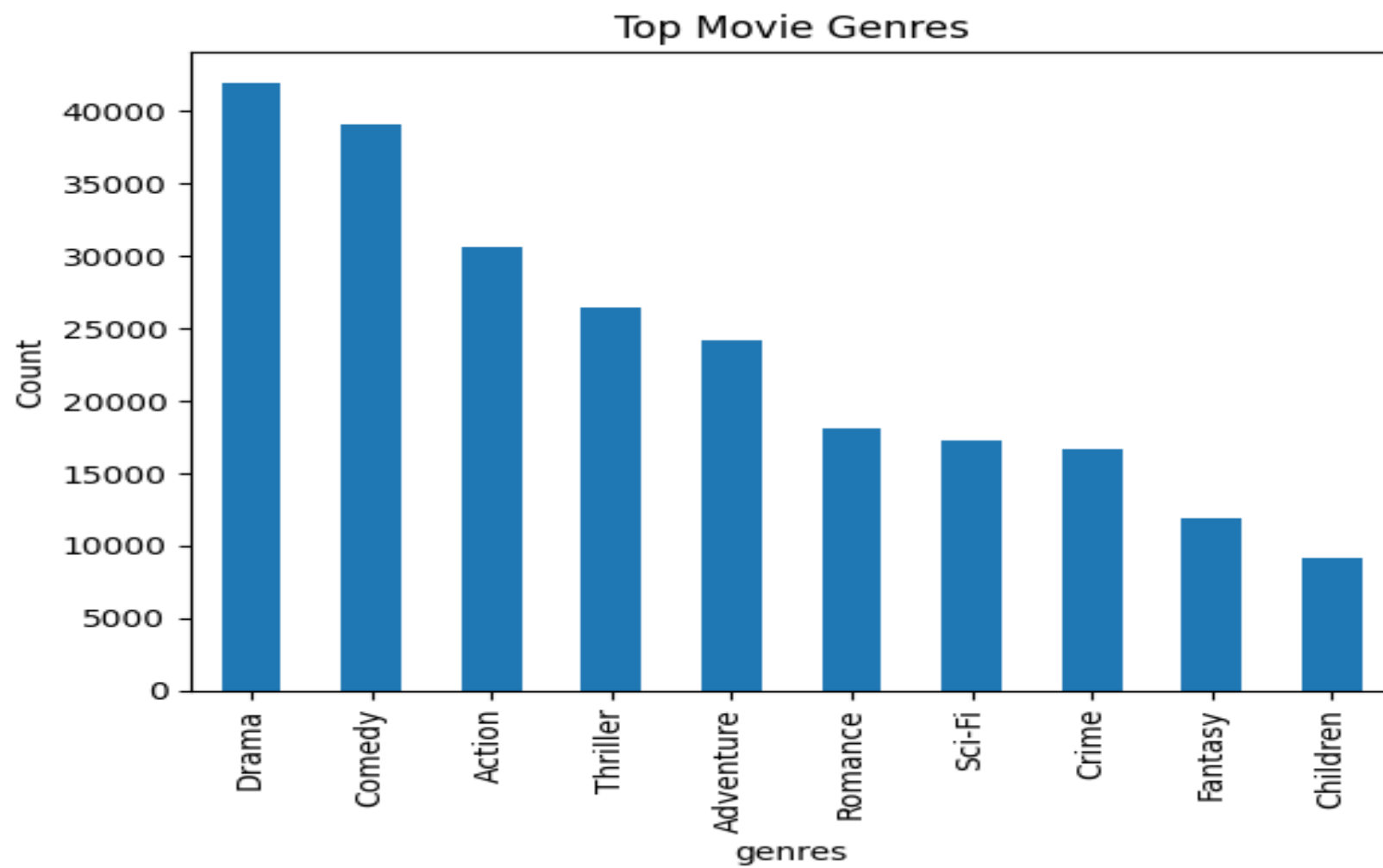
- ▶ MovieLens Latest Small Dataset
- ▶ Contains:
 - Movies (titles, genres)
 - Ratings (user preferences)
 - Tags (user description)
- ▶ Key fields: userId, movielfd, rating, genres

Data Preprocessing

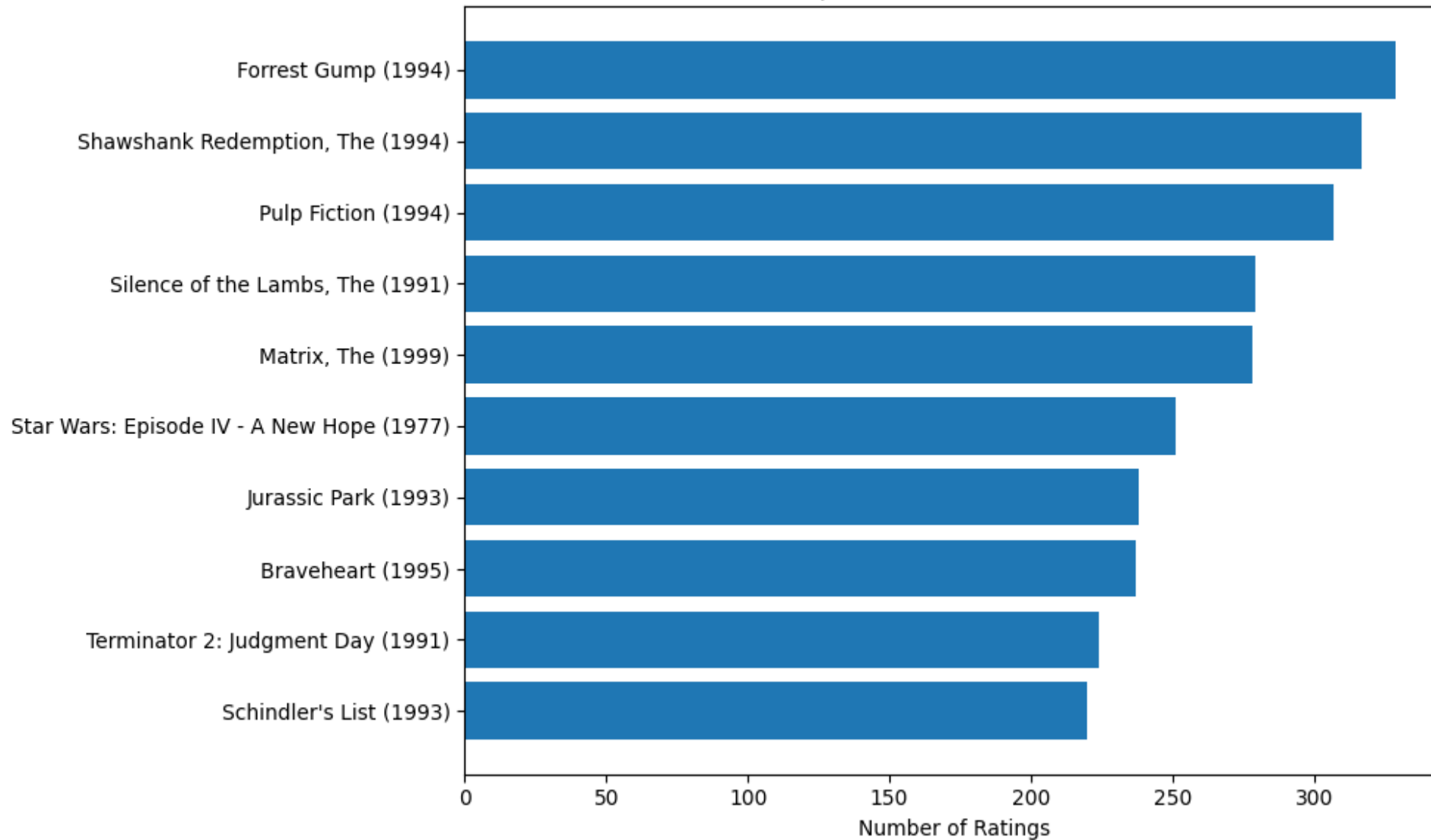
- ▶ Loaded and explored datasets
- ▶ Handled missings values
- ▶ Merged datasets for modelling
- ▶ Performed feature engineering using genres and tags

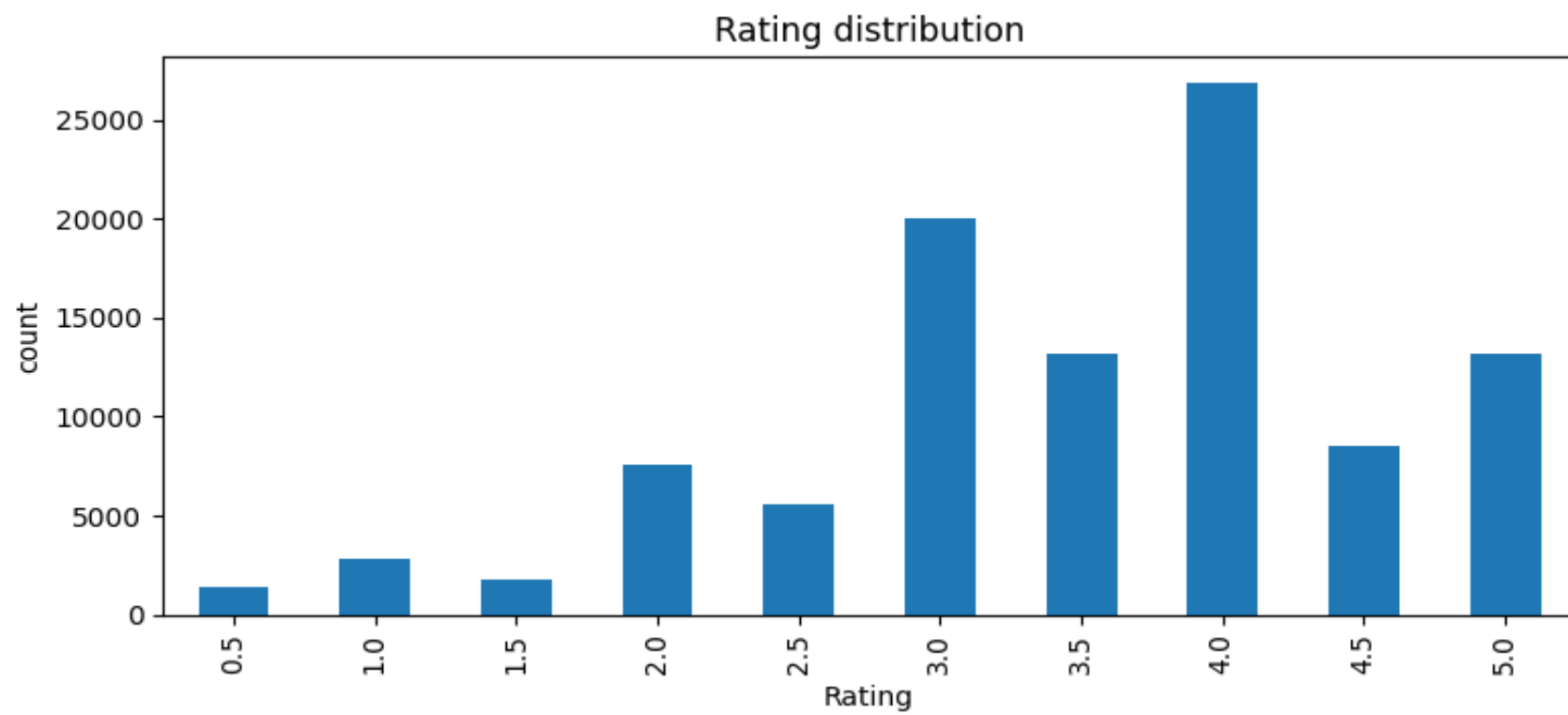
Exploratory Data Analysis

- ▶ Analyzed rating distribution
- ▶ Most popular movies
- ▶ Most rated movies
- ▶ Genre distribution insight



Top 10 Most Rated Movies

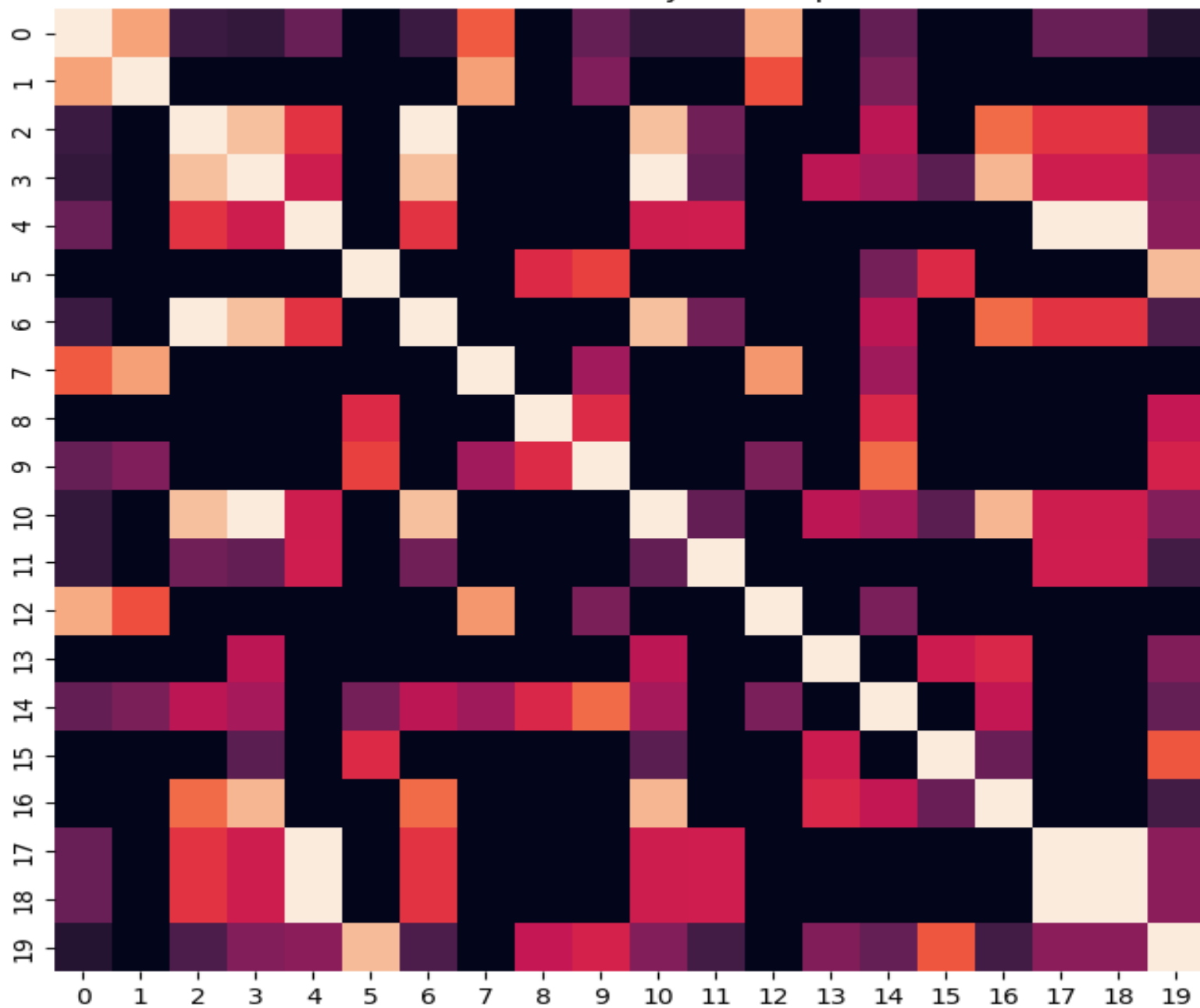




Model Development

- ▶ Content-Based Filtering
 - TF-IDF on movie genres
 - Cosine similarity
- ▶ Collaborative Filtering
 - SVD model
- ▶ Hyperparameter Tuning
 - GridSearchCv

Cosine Similarity Heatmap



Hybrid System

- ▶ Combines content-based filtering + collaborative filtering
- ▶ Improves recommendation accuracy
- ▶ Reduces limitations of individual models

Results

- ▶ Generated personalized movie recommendations
- ▶ Improved relevance of suggested movies
- ▶ Better handling of cold-start problem

Conclusion

- ▶ Successfully built a hybrid recommendation system
- ▶ Demonstrated machine learning in real-world application
- ▶ System improves movie discovery experience